

MODEL BUILDING AND CENTERING



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Applications of Hierarchical Linear Modeling

Model Building - Covariates

It is not necessary to include the same predictors for each lower-level parameter.

Level-1 Model

$$Y_{ij} = \beta_{0j} + \beta_{1j}^*(X_{ij}) + r_{ij}$$

Level-2 Model

$$\beta_{0j} = \gamma_{00} + \gamma_{01}W_j + u_{0j}$$

$$\beta_{1j} = \gamma_{10} + u_{1j}$$

Model Building – Random Effects

When building an HLM, at least one random effect needs to be included at each higher level. This will typically be a random effect for the intercept at each level.

You do not need a random effect for every covariate.

Level-1 Model

$$Y_{ij} = \beta_{0j} + \beta_{1j}^*(X_{ij}) + r_{ij}$$

Level-2 Model

$$\beta_{0j} = \gamma_{00} + u_{0j}$$

$$\beta_{1j} = \gamma_{10}$$

Model Building – Random Effects

Each random effect added to a model requires exponentially more computation, due to estimation of the covariances.

With large datasets or models, it is best to be selective in including them.

If you find that the variance of a random effect (τ) is not statistically significant, you can typically remove it from a model.

Reliability

Intuitively, reliability is how well we can tell units apart on the basis of a parameter.

$$\lambda_{0j} = \frac{Var(\beta_{0j})}{Var(\bar{Y}_{\cdot j})} = \frac{\tau_{00}}{\tau_{00} + (V_{00j})} = \frac{\tau_{00}}{\tau_{00} + (\frac{\sigma^2}{\sqrt{n_j}})}$$

$$\frac{\text{Parameter Variance}}{\text{Parameter Variance} + \text{Error Variance}}$$

$$\frac{\text{True Variance}}{\text{Observed Variance}}$$

Reliability

You have a reliability for each random effect

$$\lambda_{1j} = \frac{\text{Var}(\beta_{1j})}{\text{Var}(\bar{Y}_{\cdot j})} = \frac{\tau_{11}}{\tau_{11} + (V_{11j})}$$

Reliability

You have a reliability for each random effect

$$\lambda_{1j} = \frac{Var(\beta_{1j})}{Var(\bar{Y}_{\cdot j})} = \frac{\tau_{11}}{\tau_{11} + \frac{\sigma^2}{\sqrt{n_j}}}$$

Reliability is high when:

- The parameter varies a lot
- n_j is high

EB Estimation

By default, our intercept and coefficient estimates are based on Empirical Bayes (EB) estimation rather than OLS averages.

A key idea here is that they are reliability-weighted estimates.

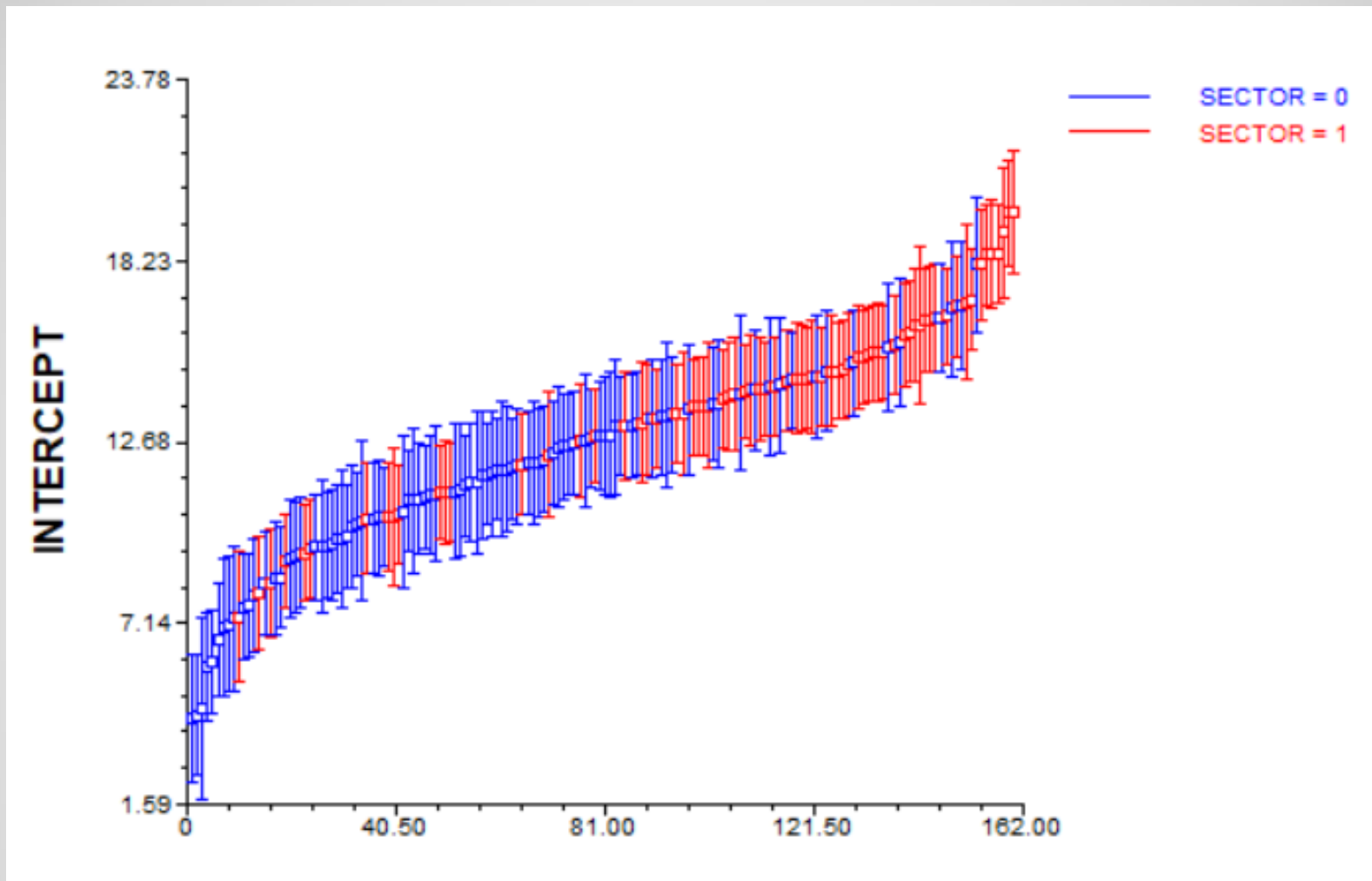
$$\beta_{0j}^* = \lambda_j \bar{Y}_{\cdot j} + (1 - \lambda_j) \hat{\gamma}_{00}$$

EB Estimation

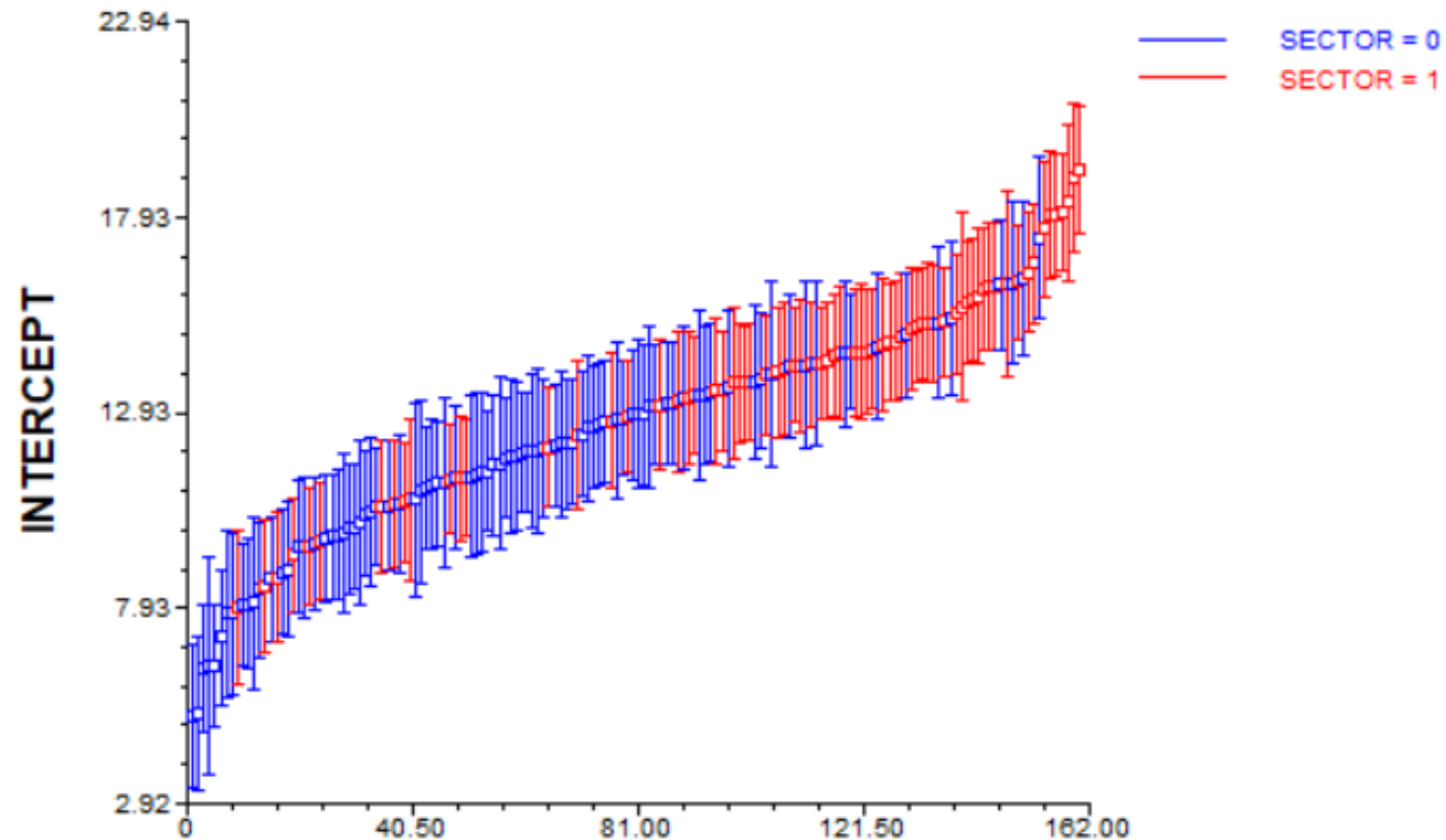
When reliability is high, the EB estimate for unit j will be close to the observed $\bar{Y}_{.j}$

When reliability is low, the EB estimate will be closer to the grand mean $\hat{\gamma}_{00}$

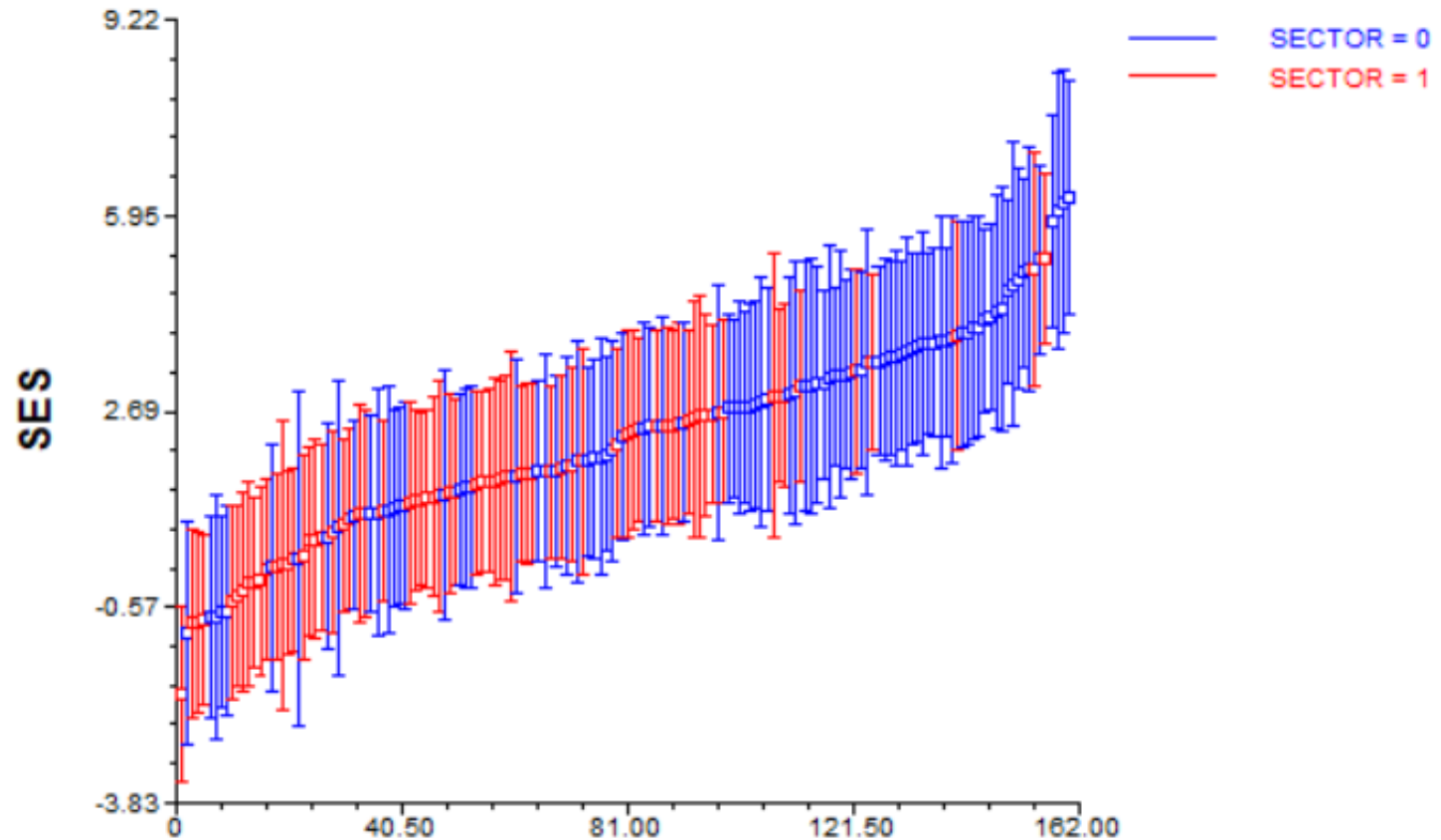
In general, these EB estimates will be closer to the true (unobserved) parameter than just using the observed means.



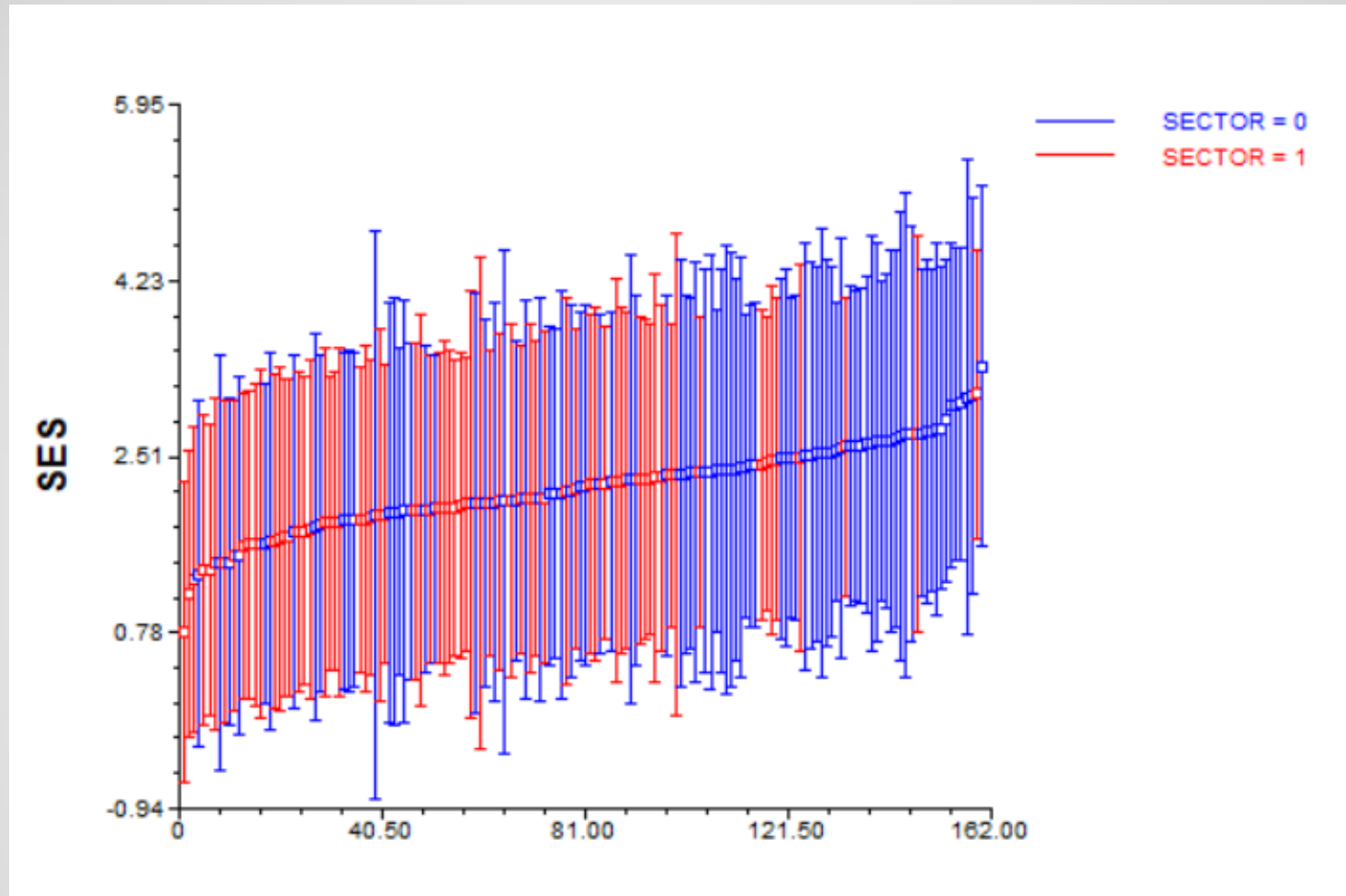
OLS Point Estimates and CIs with $\lambda_{\beta_0} = .908$



EB Point Estimates and Cis with $\lambda_{\beta_0} = .908$



OLS Point Estimates and CIs with $\lambda_{\beta_1} = .260$



EB Point Estimates and CIs with $\lambda_{\beta_1} = .260$

Centering

Centering predictors in strictly-linear OLS models only changes the intercept. They do not change coefficient estimates.

But in HLMs, centering decisions are more complicated. There are three typical options:

- Natural metric (uncentered)
- Grand-mean centered
- Group-mean centered

Natural Metric (Uncentered)

Level-1 Model

$$Y_{ij} = \beta_{0j} + \beta_{1j}^*(X_{ij}) + r_{ij}$$

Level-2 Model

$$\beta_{0j} = \gamma_{00} + u_{0j}$$

$$\beta_{1j} = \gamma_{10}$$

$$\beta_{0j} = E(Y_{ij} | X_{ij} = 0)$$

β_{1j} is the within-group effect of X , adjusted for population-level differences in X

Grand-Mean Centering

Level-1 Model

$$Y_{ij} = \beta_{0j} + \beta_{1j} * (X_{ij} - \bar{X}_{..}) + r_{ij}$$

Level-2 Model

$$\beta_{0j} = \gamma_{00} + u_{0j}$$

$$\beta_{1j} = \gamma_{10}$$

$$\beta_{0j} = \mu_{Y_j} - \beta_{1j} * (X_{ij} - \bar{X}_{..})$$

β_{0j} is the adjusted mean Y for the average unit i in the population within group j . It is adjusted for differences in X across units.

β_{1j} is the within-group effect of X, adjusted for population-level differences in X

Group-Mean Centering

Level-1 Model

$$Y_{ij} = \beta_{0j} + \beta_{1j} * (X_{ij} - \bar{X}_{.j}) + r_{ij}$$

Level-2 Model

$$\beta_{0j} = \gamma_{00} + u_{0j}$$

$$\beta_{1j} = \gamma_{10}$$

$$\beta_{0j} = \mu_{Y_j}$$

β_{0j} is the unadjusted mean of Y for unit j

β_{1j} is the unadjusted within-group effect of X (β_w)

Other options

You can center around any point of theoretical interest, as long as the point is within the range of the covariate data.

$$\beta_{0j} = E(Y_{ij} \mid X_{ij} = \text{chosen centering point})$$

Examples:

- Known population mean
- Important time point
- Salient predictor values (e.g. 18 years of age)

Centering

Centering does not change the findings for a specific research question!

Changing the centering gives you the answers to different questions.

